

Common Pediatric Infectious Diseases (Part Two): Common Clinical Syndromes

Lecture L08 · Dr. Alanoud Aljarbou, Pediatric Infectious Diseases Consultant · Summary built from the slides and all annotated notes (Notes done by Raghad Alshakhs; team leader Leena Alhadidi)

Plain text = slide content

📝 = handwritten or typed student notes on the slides / mind-map

⚠️ = the source contradicts itself or looks wrong, see the last page

Clinical syndromes covered	For each syndrome you need to know
• Pharyngitis • Acute otitis media (AOM) • Acute bacterial sinusitis (ABS) • Skin and soft tissue infection (SSTI) • Urinary tract infection (UTI)	• Causative organisms • Signs and symptoms • Diagnosis • Complication • Management

1. Pharyngitis

Pharyngitis: Definition, Causes & Age	
Definition • Infection of the pharynx and/or tonsils	
Causes	
Viruses (80%) 📝 <i>most common</i>	Bacteria (20%)
Rhinoviruses · Coronavirus · Adenovirus · Enteroviruses · Influenza viruses · Parainfluenza viruses · RSV · EBV · HSV	GABHS (15%) · Group C streptococci · Group G streptococci · <i>Corynebacterium diphtheriae</i> · <i>Arcanobacterium haemolyticum</i> · <i>Chlamydophila pneumoniae</i> · <i>Mycoplasma pneumoniae</i> · <i>Yersinia enterocolitica</i> · <i>Francisella tularensis</i>
• 📝 <i>Group A strep causes severe complications; its clinical presentation is scarlet fever rash and sandpaper rash</i> • 📝 <i>Exudative pharyngitis can be caused by EBV, adenovirus</i> • 📝 <i>Mind-map: age > 4 → mainly bacterial; age < 4 → mainly viral; 80% viral / 20% bacterial ⚠️1</i> • 📝 <i>Cough, rhinitis, conjunctivitis come with viral</i>	

Pharyngitis: Diagnostic Tests		
Test	Sensitivity %	Specificity %
Rapid antigen detection test (RADT)	70–97 📝 <i>low sensitivity</i>	> 95
Throat culture	97	99
• 📝 <i>↑ Specific → ruling IN the disease; ↑ Sensitive → ruling OUT the disease (used as screening tool)</i> • 📝 <i>RADT: ↑ specific but not sensitive → not so good for screening, so if –ve wait for culture (“–ve? you need to repeat the culture”)</i> • 📝 <i>Throat culture: ↑ specific, ↑ sensitive. Problem: it takes 2–3 days, but we have 9 days to prevent complication</i> • 📝 <i>PCR is expensive, so it's not everywhere</i> • 📝 <i>Mind-map: 1) RADT: (–) → repeat culture, (+) → treat. 2) Throat culture: (–) → symptomatic, (+) → amoxicillin</i>		

Pharyngitis: Clinical Manifestations (Viral vs Bacterial)		
Feature	Viral	Bacterial
Age	All ages	Peak: 5–11 yr

Pharyngitis: Modified Centor Criteria

Criteria	Point
Temperature > 38 °C	1
Absence of cough	1
Swollen, tender anterior cervical nodes 📌 <i>unilateral</i>	1
Tonsillar swelling or exudate	1
Age 3–14 years / 15–44 years / ≥ 45 years	1 / 0 / –1

📌 *also if it's winter season*

Score	Risk of streptococcal infection
≤ 0	1%–2.5%
1	5%–10%
2	11%–17%
3	28%–35%
≥ 4	51%–53% 📌 <i>mostly bacterial</i>

McIsaac et al. JAMA 2004; 291:1587–95

Pharyngitis: Should We Treat GAS? & Clinical Approach

Should we treat group A streptococcal pharyngitis?

- 📌 *If the patient looks well, we need to prevent **rheumatic fever**, so we have a **window of 9 days**. Do a RADT: **Positive** → **start treatment**; **Negative** → **wait for culture***
- 📌 *If the facility doesn't have RADT/culture → go to clinical diagnosis, epidemiology, and patient manifestations*
- 📌 *Mind-map: we need to treat bacterial to avoid complications like rheumatic heart disease, which are caused by group A strep. ("Glomerulonephritis" is crossed out with the Arabic note: we can't prevent it even if they took an antibiotic) **Δ2***

Clinical approach (flowchart)

- Clinical and epidemiological features**
 - Suggestive of **viral** infection → **symptomatic therapy** 📌 *no need for test*

Onset	Variable	Sudden 📌 <i>severe fever</i>
Sore throat	+	+++
Fever	Variable	+++
Abdominal pain	±	+, ++
Cough / Rhinitis / Hoarseness / Conjunctivitis / Diarrhea	+	– (each)
Other symptoms	Myalgia, arthralgia	Headache
Tonsillar hypertrophy	Variable	+++
Exudate	±	+++ 📌 <i>more</i>
Palatal petechiae	±	+
Cervical nodes	Bilateral, nontender	Unilateral, tender 📌 <i>with group A strep</i>
Rash	Characteristic exanthems	Scarlet fever rash 📌 <i>with group A strep</i>

Pharyngitis: Complications & Treatment

Complications of GAS pharyngitis

Suppurative	Nonsuppurative
Retropharyngeal abscess · Peritonsillar abscess · Cervical adenitis · Otitis media · Sinusitis · Mastoiditis · Bacteremia leading to metastatic infection	Acute rheumatic fever · Acute glomerulonephritis · Reactive arthritis (?) · Toxin-mediated: streptococcal toxic shock syndrome 📌 <i>secondary to toxin</i>

📌 *Mind-map complications: acute rheumatic fever, glomerulonephritis (non-suppurative)*

Treatment

- Penicillin V** 📌 *Group A are highly susceptible to it*
- Alternative therapy: **Amoxicillin** 📌 *Case 2: 10 days amoxicillin*
- Beta-lactam allergic patients: **Macrolide or Clindamycin**
 - 📌 *Be careful if the patient has severe infection or disseminated disease: need to have a sensitivity to them*

- Suggestive of **bacterial** infection → one of:
 - Molecular test 📌 *PCR, but expensive*: positive → antimicrobial; negative → symptomatic
 - Rapid antigen detection test 📌 *low sensitivity*: positive → antimicrobial; **negative** → **throat culture**
 - Throat culture: positive → antimicrobial therapy; negative → symptomatic therapy

Pharyngitis: Lecture Cases 1 & 2

Case 1. 3-year-old with low grade fever, runny nose, cough and sore throat. 📌 → *most likely viral (more of viral; ≤ 3 most likely viral)*

- 📌 Tx → *supportive*
- 📌 Written beside the Case 1 photo: "swelling in the tonsils + headache + unilateral tenderness in cervical LN" **Δ3**



Case 1 image (slide 6)

Case 2. 6-year-old with high fever and severe sore throat 📌 *and absence of cough.* 📌 → *bacterial, > 3 y Δ1*

- 📌 Group A strep → *treat to prevent complications of rheumatic fever and glomerulonephritis Δ2*
- 📌 Scarlet fever
- 📌 Give antibiotic → *10 days amoxicillin*
- 📌 Exudate (labelled on the tonsils)



Case 2 image (slide 6)

2. Acute Otitis Media (AOM) 📌 *caused by bacteria*

AOM: Case 3 & Who to Treat

Case 3. 6-month-old infant with fever, vomiting and excessive crying. Ear examination showed the following:

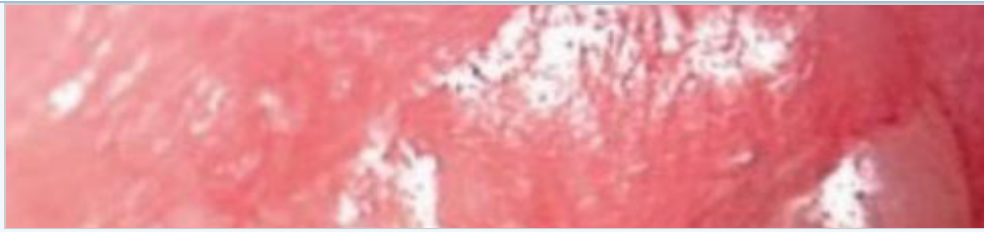
AOM: Clinical Presentation & Complications

Symptoms

- Ear pain (**ear tugging** in infant)
- Irritability / crying
- Fever
- Ear discharge 📌 *with or without perforation*
- 📌 *diarrhea and sometimes vomiting*

Signs

- Dullness of TM
- Absence of light reflex
- **Bulging TM** 📌 *hallmark*
- Limited or absent mobility
- Air fluid level
- Otorrhea



Case 3 image (slide 16)

- 📌 *Tympanic membrane bulging → only with acute suppurative otitis media*
- 📌 *Redness → can happen with other than acute suppurative otitis media*

Treatment decision 📌 (typed notes)

- > 2 Y with **mild** symptoms → **can wait**
- > 2 Y with **severe**: high grade fever, bilateral disease, severe otalgia → **treat!**
- < 2 Y, especially < 6 M → **have to treat**; 6 M → antibiotics
- 📌 *Antibiotic because the most common causes are bacteria. Although it's self-limiting, we need treatment if < 2 y, especially 6 m, because of ↑ rate of complications*
- 📌 *Approach: symptoms resolved within 48–72 h → no need for antibiotic, only supportive*
- 📌 *Symptoms persist → start antibiotic. If < 2 y we give **10 days** to prevent treatment failure ("symptoms improved by day 5"), or if the patient is really sick give treatment immediately. > 2 y: **5 days** treatment*
- 📌 *Mind-map: age < 2 year → treat; age ≥ 2 y: if symptoms resolve in first 48–72 h no need for antibiotic, just supportive, but if symptoms persist start antibiotic. Duration: > 2 y → 5 d; < 2 y → 10 d*

AOM: Pathophysiology

Viral URTI →

↓ Mucociliary defense · ↑ Bacterial colonization · ↑ Bacterial adhesion · Altered immune response

→ **Ascending bacterial (± viral) infection via ET** → **AOM**

- Viral URTI → **secretions and edema** 📌 *mainly* → **ET dysfunction** 📌 *hallmark*; ↑ pressure; leads to -ve air pressure in ear → AOM
- ET dysfunction → **negative ME pressure** → reflux of nasopharynx microbes into ME → ascending infection
- Viral URTI → **nose blowing, sniffing** → reflux of nasopharynx microbes into ME
- ET dysfunction → **mucus accumulation**: medium for bacterial growth → AOM

Complications of AOM

- Tympanic membrane perforation
- Conductive hearing loss
- Mastoiditis
- Labyrinthitis
- Brain abscess
- Epidural abscess
- Otic hydrocephalus (increased ICP without hydrocephalus)

📌 *Mind-map complications: TM perforation, conductive hearing loss, mastoiditis, brain abscess*

AOM: Etiology & Antibiotic Choice

Etiology

- **Bacteria (60%–80%)***
 - **S. pneumoniae: 30%** 📌 *although we have a vaccine, there are subtypes*
 - H. influenzae: 20% 📌 *"nontypeable, mostly type B" ⚠️4*
 - M. catarrhalis: 20%
- **Viruses (20%–40%)**
- * **Spontaneous resolution: 50%** 📌 *> 2 y pt*

📌 Typed notes on the etiology slide

- 📌 *Causative organisms for URI "bacterial": S. pneumoniae (marked with asterisk), H. influenzae type B, M. catarrhalis (Moraxella)*
- 📌 *Why is S. pneumoniae still the most common cause of AOM although we have a vaccine? Because not all subtypes are covered by the vaccine, only those that cause invasive disease are covered*
- 📌 *H. influenzae → usually resistant to amoxicillin, that's why we go with amox-clav; but nowadays H. influenzae is mostly eradicated by vaccines. "If PT have symptoms e.g. current lymphadenitis" (written as is)*
- 📌 *Tx → **amoxicillin, we need a high dose** ("Needs high doses")*
- 📌 *If we're afraid the patient will have resistance, or they used amoxicillin in the last month → give **amoxiclav***
- 📌 *Mind-map: Strep pneumo 30% (most); if they have **conjunctivitis** → **mostly H. influenzae**; if we suspect H. influenzae we give amoxi-clav because it is resistant to amoxicillin*

- Adenoids: bacterial reservoir? → ascending infection

📌 Student-added ChatGPT explanation (screenshot on slide 18)

1. Viral URTI: often the initial event; causes inflammation and irritation of the respiratory tract
2. Impaired mucociliary defense: less clearance of pathogens from the upper respiratory tract
3. Eustachian tube dysfunction (starred): secretion and edema (swelling) block the ET, which normally ventilates the middle ear, so negative pressure builds up in the middle ear
4. Nose blowing and sniffing push nasopharyngeal microbes into the middle ear via the ET
5. Bacterial colonization and adhesion → ascending bacterial (± viral) infection via the ET
6. Mucus accumulation in the middle ear → medium for bacterial growth → ear pain and inflammation
7. Adenoids can act as a bacterial reservoir

AOM: Clinical Approach (Flowchart: clinical and epidemiological features of AOM)

Step 1: Treat now or observe?

- **Age < 2 yr** → **Treat** 📌 *tx 10 d*
 - **Age ≥ 2 yr** 📌 *wait, if not improved treat for 5 days* → is the child **ill and toxic, T ≥ 39 °C, severe otalgia** (typed addition: + **bilateral OM**)?
 - Yes → **Treat**
 - No 📌 *wait* → **symptomatic therapy, reassess after 48 hr** → improved → continue symptomatic therapy; not improved → **Treat**
- 5 days non-inferior to 7 days in children > 2 years** (slide side note)

Step 2: Which antibiotic? ("Treat" box)

- Received **amoxicillin within 30 days** OR **concurrent purulent conjunctivitis?** 📌 *usually due to H. influenzae resistance*
 - **No** → **high dose amoxicillin** → improved → continue amoxicillin **7 days** for uncomplicated cases > 2 yr and **10 days** for other cases; not improved → high dose amox/clav
 - **Yes** → **high dose amox/clav** 📌 *if the patient took amoxicillin alone, treatment fails because there is resistance* → improved → continue same therapy for 10 days
 - Not improved on amox/clav → **IV ceftriaxone 50 mg/d daily for 3 days** ⚠️ → if no improvement → **tympanocentesis & adjust therapy according to C/S** 📌 *culture usually for immunocompromised pt*

3. Acute Bacterial Sinusitis (ABS)

ABS: Background & Pathogenesis

- 📌 *AOM → 90% preceded by URI; ABS → 100% preceded by URI (mind-map: "100% preceded by URTI")*
- 📌 *High-risk patient: damaged cilia → cystic fibrosis*
- 📌 *Mind-map: child < 4 y does not have sinusitis because the sinuses are still underdeveloped*

Pathogenesis

- The respiratory mucosa in the paranasal sinus is continuous with the nasal mucosa.
Most cases of ABS followed URTI

ABS: Clinical Manifestations, Etiology & Diagnosis

Clinical manifestations (3 types)

Persistent	Nasal discharge and/or cough for > 10 days and not improving
Severe	High fever (≥ 39 °C) and purulent nasal discharge together for > 3 days
Worsening	Resolving upper respiratory symptoms, then worsening on day 6 or 7 with new or recurrent fever 📌 <i>resolve then recur again</i> or exacerbation of nasal symptoms and/or cough

Etiology

- **Streptococcus pneumoniae: 30–40%**

- Three key elements to normal physiology of the sinuses:

1. **Patency of the ostia** (can be obstructed following URTI) *due to a lot of secretions*
2. **Function of ciliary apparatus** (if impaired, bacteria multiply) *so it will move the secretion down to the nose; some patients don't have this function due to congenital [causes], cystic fibrosis, acute URI*
3. **Quality of the secretions** (over-production occurs with URTI)

Anatomy figure (Fig. 32.3, coronal section)

- 1 maxillary sinus · 2 ethmoidal bursa · 3 ethmoid cells · 4 frontal sinus · 5 uncinate process · 6 middle turbinate · 7 inferior turbinate · 8 nasal septum · **9 ostiomeatal complex** (blue area)
- *Ostiomeatal complex gets obstructed by thick secretion due to URI → good media for bacteria to grow, so it causes sinusitis; fluid accumulation (the next handwritten line is cut off at the slide edge in the source)*

- Haemophilus influenzae: 20%
- Moraxella catarrhalis: 20%
- Viruses: 10%
- Staphylococci & anaerobes are uncommon in ABS *in immunocompromised*
- *also fungal in immunocompromised*

Diagnosis

- In uncomplicated sinusitis **no imaging indicated** *usually the only finding is mucosal thickening of the sinuses, and that could happen with acute viral infection; clinical is enough because findings are not specific (we would find URI findings)*
- **Sinus aspiration** indicated in special situations only *if not responding to tx or in immunocompromised pt*
- *"first wait 48 h then give tx" ⚠6*

ABS: Who to Treat & Clinical Approach (Flowchart)

Who to treat? *(mind-map)*

- *Severe or worsening symptoms, OR*
- *Has another bacterial infection (AOM / pneumonia), OR*
- *Ciliary defect: cystic fibrosis, ciliary dysfunction*
- **Treat: high-dose amoxi-clav for 7–10 d**
- *Complication: **brain abscess** (only in the mind-map; the slides have no ABS complications slide)*

Flowchart

- Any of: severe or worsening symptoms · concurrent bacterial infection (e.g. pneumonia, AOM) · suspected complications · underlying conditions (e.g. BA, CF, ID)?
 - **Yes → Treat**
 - **No → outpatient observation for 3 days** → improving → continue symptomatic therapy; not improving → Treat
- Treat: child in **daycare** OR has **severe symptoms** OR **received amoxicillin within 30 days?**
 - No → **high dose amoxicillin** → improving → continue same therapy to complete **7 days beyond resolution of symptoms** *7–10*; not improving → high dose amox/clav
 - Yes → **high dose amox/clav** → improving → same as above; not improving → **sinus imaging · sinus aspiration** *fungal in immunocompromised* · **adjust therapy according to C/S**

Student-added ChatGPT explanation (slide 29) says the same steps; "BA" = bronchial asthma, "ID" = immunodeficiency; it reads "CE" as chronic ear infection, but the flowchart box says "CF" (cystic fibrosis). If improving, continue until 7 days after symptom resolution (total 7–10 days, underlined in the notes).

4. Skin and Soft Tissue Infections (SSTI)

- *Causative organisms: Staph and group A strep*

Case 4 & Impetigo

Case 4. 6-year-old boy with redness and swelling of the chin with golden crusts. What is your management if he looks well and afebrile? If ill and febrile?

- 📌 *Well and afebrile* → **give topical antibiotic**
- 📌 *Ill and febrile* → **systemic tx**
- 📌 *Golden crust, honey lesions, highly contagious; impetigo crust* → **group strep**



Case 4 image (slide 30)



Fig. 68.1 bullous impetigo, *S. aureus*

Impetigo

- Localized, **highly contagious**
- Caused by ***S. aureus* and GAS**
- Two forms: **bullous** 📌 *due to toxin; S. aureus* and **nonbullous** 📌 *group A*
- 📌 *Staph: due to toxin, not the organism. Bullous is secondary to staph; heals with no scar*
- Children **2–5 years** are affected most often
- Fig. 68.1: scattered pustules on the abdomen and thighs of an infant with bullous impetigo (*S. aureus*); one lesion ruptured, leaving an erosion with a collarette of scale

Management

- **Topical antibiotics** (e.g. **mupirocin**) are sometimes effective for mild cases 📌 *if afebrile*
- **Narrow-spectrum systemic antibiotics** (e.g. **flucloxacillin**) are generally needed for more severe infections 📌 *if pt febrile*

Folliculitis, Furuncles & Carbuncles

Folliculitis 📌 *only epidermis*

- Superficial infection of the hair follicle: discrete, **2–5 mm papules and pustules on an erythematous base**
- Single or grouped; a **hair shaft often seen in the center**
- ***S. aureus*** is the predominant pathogen 📌 *staph, group A*
- 📌 *More superficial than furuncle and more spread*

Ecthyma, Ecthyma Gangrenosum, Erysipelas, Cellulitis

Ecthyma 📌 *ulceration*

- Rare, **deep ulcerative** infection penetrating to the **dermis**
- Most commonly ***S. pyogenes***
- **Systemic antibiotic** effective against streptococci

Ecthyma gangrenosum

- **Serious**, characteristic lesion with gram-negative organisms, especially ***Pseudomonas aeruginosa*** 📌 *2nd to toxin*
- Prompt antibiotic effective against *P. aeruginosa* or other likely organisms
- 📌 *Necrotic center, more serious, gram -ve, Pseudomonas especially in immunocompromised* → **start anti-pseudomonal and take blood culture**
- Fig. 68.8: *P. aeruginosa* septicemia and ecthyma gangrenosum in a boy with severe neutropenia; fatal case in an infant with leukemia; H&E shows superficial necrosis with bland ischemic necrosis beneath; Gram stain of bullous fluid shows dense gram-negative bacilli with rare inflammatory cells

Erysipelas

- Superficial infection of the **upper dermis and the lymphatic system** 📌 *superficial, demarcated (following lymphatic chain)*
- Mostly **GAS**; occasionally group B, C, G strep 📌 *group A more than staph*
- Treatment: **oral anti-streptococcal agent for 7–14 days** ⚠️

Cellulitis 📌 *deep*

- Acute infection of the **dermis and subcutaneous tissues**: edema, warmth, erythema, tenderness
- Most common: ***S. pyogenes* and *S. aureus***
- Uncomplicated: antimicrobials targeting **streptococci and staphylococci**
- 📌 *Cellulitis and erysipelas: both need oral AB for 5–7 days* ⚠️

The SSTI photo slide also shows **erythrasma**, with no text about it anywhere in the lecture.

Folliculitis management

- Simple bacterial folliculitis often **resolves spontaneously without scarring**
- Mild infection: topical antibiotic cleansers (chlorhexidine), topical antibacterials (mupirocin, clindamycin), and benzoyl peroxide are usually effective 📌 *topical antiseptic and antibiotic*

Furuncles and carbuncles 📌 *epidermis + dermis*

- Furuncle (boil)**: infection of the hair follicle; unlike folliculitis (stays in the epidermis), **extends deep into the dermis** 📌 *deep*
- Carbuncle**: painful infection of an **aggregate of contiguous follicles**, with **multiple drainage points** and inflammation of the surrounding connective tissue
- Cause: almost always **S. aureus** (Fig. 68.6 furuncle due to S. aureus)
- Treatment: frequent **hot, moist compress** to promote drainage
- Large furuncles and most carbuncles require surgical drainage**

5. Urinary Tract Infection (UTI)

UTI: Case 5, Definition & Pathogenesis

Case 5. 9-month-old infant with fever for three days; on examination: well, T 38.9 °C, no focus. What is your next step? 📌 *urine analysis*

- 📌 *Usually in infants the only finding is fever; there is no focus*

Definition 📌 *imp*

- Defined as **significant bacteriuria** of urinary pathogen in a **symptomatic patient**
- 📌 *Mind-map (Arabic): if they have no symptoms we don't treat, except pregnant women or before procedures*

Pathogenesis ("for your knowledge")

- Colonization of periurethral mucosa with gastrointestinal bacteria → ascend into the bladder, ureters, and kidneys
- Bacterial virulence factors 📌 *like Klebsiella: attaches easily to the ureter and travels to the kidney*
- Host risk factors: obstruction to urine flow with urosthesis (calculi, tumor, **constipation**), **neurogenic bladder**, **vesicoureteral reflux** 📌 *obstruction*
- Circumcision reduces UTI in boys < 3/12 (under 3 months) by almost 8-fold** 📌 *"because of short urethra" Δ8*

UTI: Diagnosis (Microscopy & Urinalysis)

Urine microscopy (pyuria)

- ≥ **5 WBC/HPF** in a centrifuged sample, or
- ≥ **10 WBC/mm³** in an uncentrifuged sample

Urine analysis: useful screening tool 📌 *↑ sensitive; if -ve the patient doesn't have UTI*

Test	Sensitivity (range) %	Specificity (range) %
Leukocyte esterase 📌 <i>WBC degradation</i>	83 (67–94)	78 (64–92)
Nitrite	53 (15–82)	98 (90–100)
LE or nitrite positive	93 (90–100)	72 (58–91)
Microscopy, WBCs	73 (32–100)	81 (45–98)
Microscopy, bacteria	81 (16–99)	83 (11–100)

📌 *We need all to be -ve (to rule out). Mind-map: urine analysis → if (-) rule out; urine culture → gold standard*

UTI: Clinical Features by Age

Age group	Most common	Intermediate	Least common
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UTI: Etiology

- **E. coli: 70–80%** 📌 *from anal area*
- Klebsiella · Proteus · Pseudomonas · Enterococcus 📌 *"(+ GPS" (as written) · Enterobacter*

Other causes

- Candida 📌 *mind-map: → urinary catheter; Tx: remove catheter only*
- Group B streptococci (**neonates**) 📌 *"GPS → in neonate" (as written)*
- Staphylococci (rarely aureus; **saprophyticus in adolescent girl**) 📌 *S. saprophyticus in young female (+)ve*
- Adenovirus (**hemorrhagic cystitis**) 📌 *in immunocompromised*
- TB (**sterile pyuria**) 📌 *↑ WBC in urine; there is pus, however the culture is –ve*

Local (Saudi) data

- **E. coli and K. pneumoniae** are the most frequent uropathogens in Saudi children
- Very high resistance to **ampicillin (~70%)** and **co-trimoxazole (~60%)**
- Most isolates highly susceptible to a **third-generation cephalosporin and aminoglycoside** 📌 *in KSA due to resistance*

Infants < 3 months	Fever, vomiting, lethargy, irritability	Poor feeding, failure to thrive	Abdominal pain, jaundice, haematuria, offensive urine
≥ 3 months, preverbal	Fever	Abdominal pain, loin tenderness, vomiting, poor feeding	Lethargy, irritability, haematuria, offensive urine, failure to thrive
≥ 3 months, verbal	Frequency, dysuria	Dysfunctional voiding, changes to continence, abdominal pain, loin tenderness	Fever, malaise, vomiting, haematuria, offensive urine, cloudy urine

Lower UTI (cystitis) 📌 <i>infection only in bladder</i>	Upper UTI (pyelonephritis) 📌 <i>infection in kidney</i>
Dysuria · urgency · frequency · suprapubic pain · hematuria	Fever (highlighted) · shivering · vomiting, nausea · diarrhea · abdominal pain / flank · lower UTI symptoms
📌 <i>Tx: less than 5 d (mind-map: 3–5 days) ⚠️9</i>	📌 <i>Tx: 7–14 d</i>

UTI: Urine Culture & Collection Methods

Method	Recommendation	Cut-off for significant bacteriuria
Transurethral bladder catheterization (in/out) 📌 <i>useful in infant</i>	First choice for infants and non-toilet-trained children	> 50,000 CFU/mL , OR > 10,000 CFU/mL if fever and pyuria
Clean-voided specimen 📌 <i>midstream sample</i>	First choice for toilet-trained children; second choice for non-toilet-trained children who cannot undergo transurethral BC (e.g. bleeding tendency)	> 100,000 CFU/mL , OR > 10,000 CFU/mL if fever and pyuria
Suprapubic aspiration 📌 <i>in bladder</i>	Reliable in neonates; alternative for males with phimosis or females with severe labial adhesions	Any number of CFU 📌 <i>even one</i>

📌 Typed collection notes

- 📌 *Suprapubic aspiration (neonate, and neurological): significant if ANY number of CFU is present*

UTI: Imaging Studies & Timing

- **First febrile UTI in a child < 3 years** 📌 *for any anomalies → RBUS* 📌 *after acute infection, 2–6 weeks. Why? Because sometimes there is some debris that blocks the ureter during infection, so it gives a false +ve → abnormal result → VCUG*
- **Child at any age with:** recurrent UTI* 📌 *more than 2 UTI in 1 year ⚠️12*, complicated UTI**, febrile non-E. coli UTI, OR family history of VUR → **RBUS & VCUG** → abnormal result → **refer to urology or nephrology*****
- 📌 *Handwritten: 1st febrile UTI < 3 years → US after 2–6 weeks; non-[E. coli] UTI → VCUG + US. Mind-map: "you should do RBUS in < 3 y age after 6 weeks to look for anomalies" ⚠️13*
- * Recurrent UTI: two or more symptomatic episodes within 12 months
- ** Complicated UTI: critically ill (urosepsis or shock), evidence of urinary tract obstruction, or failure to respond after 48 hours of appropriate antibiotics
- *** Refer to nephrology/urology: recurrent UTIs with abnormal renal-bladder imaging, moderate to high-grade reflux (grade III to IV), congenital renal anomalies, bladder dysfunction, impaired kidney function or hypertension

Timing of RBUS

- Depends on the clinical condition
- **2 to 6 weeks after the acute phase** to reduce false-positive results from renal parenchymal inflammation

- In/out catheterization: 50,000 CFU/mL (or 10,000–50,000 range depending on guidelines); in/out done if not toilet trained
- Mid-stream urine: 100,000 CFU/mL is significant
- Bag collection: for urinalysis it rules out UTI; for urine culture do **not** use a bag culture for definitive diagnosis **Δ10**
- Colonization: from inside the human body. Contamination: from outside the body
- OSCE: 10 thousand in/out, mid-stream → urinalysis, fever

Urine bag?

- Not recommended: **contamination rate is extremely high** ✖ *urine bag, because of the risk of contamination; mind-map (Arabic): avoid it*
- Helpful mainly in children with a negative culture result to exclude UTI (slide wording) **Δ10**

UTI: Diagnostic Approach (Flowchart)

- Suspect UTI in **infants & young children with fever and no clear focus**, AND in **older children with urinary symptoms**
- Collect urine by the recommended methods (Table 1) → send urine for **dipstick, microscopy and culture**
- **Febrile child with urinary symptoms OR unwell-appearing child** → a) start empiric antibiotic while waiting for culture; b) adjust according to culture
- **Afebrile child with urinary symptoms OR well-appearing child with fever and no apparent source** ✖ *wait for culture* → check dipstick and microscopy:
 - a) Positive for any of LE, nitrite, pyuria or bacteriuria → start empiric antibiotic while waiting for culture
 - b) Negative → wait for final culture; consider repeating dipstick & microscopy after 24–48 h if fever persists
- UTI should be considered in infants and young children with fever $\geq 38^\circ\text{C}$ and **no apparent source**, especially with risk factors: **age < 12 months, female, uncircumcised boy, temperature > 39°C, fever > 48 hours**

- **Early (during the acute phase) if severely ill, complicated UTI, or evidence of obstruction**, to identify renal or perirenal abscesses and pyonephrosis
- Typed notes: *exception for early US if severely ill or fails to improve within 48 hours of antibiotics; if post-illness RBUS is abnormal → order VCUG*

UTI: Treatment

- **Oral antibiotics are usually appropriate**; in children aged ≥ 2 months, starting orally or parenterally is **equally effective** ✖ *mind-map: oral = IV unless toxic patient*
- **Toxic patients or unable to take orally** → parenteral antibiotics until clinically improved for **24–48 hours**
- ✖ *Mind-map: empiric amoxiclav / ceftriaxone, then depend on culture → step down. Duration 7–14 days*

Table 3: empiric antibiotics for community-acquired UTI (age ≥ 3 months to 14 years)

Setting	First line (one of)	Alternative ^a	Duration
Outpatient	Amoxicillin-clavulanic acid ✖ <i>1st choice</i> ; cephalexin, cefuroxime, cefprozil	Cefixime ✖ <i>regular dose</i>	3–7 days
Inpatient ^a	Ceftriaxone	Gentamicin ^b ± ceftriaxone ^c	7–14 days ✖ <i>pyelonephritis</i>

- Check any previous urine culture and susceptibility results and choose antibiotics accordingly
- ^a Alternative: if no improvement of fever and UTI symptoms at least 48 h after starting first choice, or first choice not suitable
- ^b Gentamicin: previous UTI caused by ESBL-producing bacteria, or recent cephalosporin exposure in the last 3 months
- ^c Ceftriaxone + aminoglycoside: critically ill, or condition worsens after first-line therapy
- ✖ *Handwritten: 1) amoxicillin-clavulanic acid; 2) "2nd or 3rd generation of cephalosporin" Δ11*

✖ Typed notes (slide 62)

- **Step 1, uncomplicated outpatient UTI (oral):** first line choice 1 amoxicillin-clavulanic acid (Augmentin); choice 2: 1st & 2nd generation cephalosporins (cephalexin, cefuroxime, cefprozil); alternative: cefixime (3rd gen oral) at regular doses if first-line fails or can't be used; duration typically 3–7 days for uncomplicated cystitis, extended to 10 days depending on severity and age
- **Step 2, complicated / inpatient UTI or pyelonephritis (parenteral):** when toxic/ill, vomiting, unable to tolerate oral, or pyelonephritis; primary IV choice ceftriaxone; alternative / resistant coverage gentamicin (prior ESBL or cephalosporin within 3 months); critically ill: ceftriaxone + gentamicin; duration 7–14 days total for pyelonephritis and systemic upper UTI

Summary

Syndrome	Main organisms	Hallmark / key features	Diagnosis	Management	Complications
Pharyngitis	Viruses 80%; bacteria 20% (GABHS 15%)	GAS: sudden high fever, severe sore throat, exudate, unilateral tender nodes, no cough, scarlet fever rash; peak 5–11 y	Modified Centor; RADT (specific, low sensitivity) → if –ve throat culture (gold); PCR expensive	Viral: symptomatic. GAS: penicillin V; alt. amoxicillin (10 days per notes); allergy: macrolide or clindamycin. 9-day window to prevent rheumatic fever	Suppurative (abscesses, adenitis, OM, sinusitis, mastoiditis, bacteremia); nonsuppurative (ARF, AGN, reactive arthritis?, STSS)
AOM	Bacteria 60–80%: <i>S. pneumoniae</i> 30%, <i>H. influenzae</i> 20%, <i>M. catarrhalis</i> 20%; viruses 20–40%	Ear pain/tugging, crying, fever; bulging TM (hallmark); ET dysfunction is the key mechanism	Clinical (ear examination); tympanocentesis + C/S if failing	< 2 y: treat 10 d. ≥ 2 y: treat if toxic/T ≥ 39/severe otalgia (+ bilateral), else observe 48 h; 5 d. High-dose amoxicillin; amox/clav if amoxicillin in last 30 d or purulent conjunctivitis; then IV ceftriaxone × 3 d	TM perforation, conductive hearing loss, mastoiditis, labyrinthitis, brain/epidural abscess, otic hydrocephalus
ABS	<i>S. pneumoniae</i> 30–40%, <i>H. influenzae</i> 20%, <i>M. catarrhalis</i> 20%, viruses 10%	100% preceded by URTI; persistent > 10 d, severe (≥ 39 °C + purulent discharge > 3 d), or worsening day 6–7	Clinical; no imaging if uncomplicated; aspiration only in special cases	Treat if severe/worsening, concurrent bacterial infection, complications, underlying disease; else observe 3 d. High-dose amoxicillin or amox/clav; 7 d beyond resolution (7–10 d)	Brain abscess (mind-map)
SSTI	<i>S. aureus</i> , GAS (<i>Pseudomonas</i> in ecthyma gangrenosum)	Impetigo: honey crusts, 2–5 y; bullous (staph toxin) vs nonbullous (GAS)	Clinical; blood culture in ecthyma gangrenosum	Mild/afebrile: topical mupirocin; severe/febrile: systemic (flucloxacillin); furuncle/carbuncle: hot compress, drainage	Ecthyma gangrenosum = serious (immunocompromised)
UTI	<i>E. coli</i> 70–80%, <i>Klebsiella</i> , <i>Proteus</i> , <i>Pseudomonas</i> , <i>Enterococcus</i> , <i>Enterobacter</i> ; GBS in neonates	Infants: fever with no focus; older: frequency, dysuria. Upper = fever, flank pain	UA (screen) + culture (gold); catheter > 50,000, midstream > 100,000, SPA any; avoid bags	Oral = IV if ≥ 2 mo and not toxic; outpatient amox-clav (3–7 d); inpatient ceftriaxone ± gentamicin (7–14 d)	Renal/perirenal abscess, pyonephrosis; RBUS 2–6 wk after first febrile UTI < 3 y

Skin Lesions at a Glance

Lesion	Depth	Organism	Clue	Treatment
Impetigo	Superficial	<i>S. aureus</i> (bullous), GAS (nonbullous)	Golden/honey crust, highly contagious, 2–5 y	Topical mupirocin (afebrile) / systemic flucloxacillin (febrile)
Folliculitis	Epidermis only	<i>S. aureus</i>	2–5 mm papules/pustules, hair shaft in center	Often self-resolves; chlorhexidine, mupirocin, clindamycin, benzoyl peroxide
Furuncle	Deep into dermis	<i>S. aureus</i>	Boil of one follicle	Hot moist compress; drain if large
Carbuncle	Dermis + surrounding tissue	<i>S. aureus</i>	Contiguous follicles, multiple drainage points, painful	Most need surgical drainage
Ecthyma	Deep ulcer to dermis	<i>S. pyogenes</i>	Ulceration, rare	Systemic anti-streptococcal
Ecthyma gangrenosum	Necrotic	<i>P. aeruginosa</i> (gram –ve)	Necrotic center, immunocompromised/neutropenic	Prompt anti-pseudomonal + blood culture
Erysipelas	Upper dermis + lymphatics	GAS (B, C, G occasionally)	Superficial, sharply demarcated	Oral anti-streptococcal 7–14 d (notes: 5–7 d)
Cellulitis	Dermis + subcutaneous	<i>S. pyogenes</i> , <i>S. aureus</i>	Edema, warmth, erythema, tenderness	Anti-strep + anti-staph

Ultra-Quick Differentiation (Keywords from this lecture)

Keyword combination	Think of
3-year-old, low-grade fever, runny nose, cough, conjunctivitis, sore throat	Viral pharyngitis → supportive
Child > 3–4 y, sudden high fever, severe sore throat, no cough, exudate, unilateral tender nodes	GAS pharyngitis → penicillin V / amoxicillin
Sore throat + sandpaper rash	Scarlet fever (GAS)
Exudative pharyngitis, not strep	EBV, adenovirus
Centor ≥ 4	51–53% strep risk, mostly bacterial
Infant with ear tugging, crying, fever + bulging TM	AOM
AOM + purulent conjunctivitis, or amoxicillin in last 30 days	H. influenzae resistance → high-dose amox/clav
Increased ICP without hydrocephalus after AOM	Otic hydrocephalus
Nasal discharge/cough > 10 days, not improving	Persistent ABS
≥ 39 °C + purulent nasal discharge > 3 days	Severe ABS
URI improving then new fever on day 6–7	Worsening ABS
Recurrent sinusitis/otitis, damaged cilia	Cystic fibrosis / ciliary dysfunction
Golden (honey) crusts on the chin, 2–5 y	Nonbullous impetigo (GAS)
Bullous lesions in an infant, heal with no scar	Bullous impetigo (S. aureus toxin)
Papule/pustule with hair shaft in the center	Folliculitis
Multiple draining follicles, painful	Carbuncle
Necrotic black center in a neutropenic child	Ecthyma gangrenosum (Pseudomonas)
Superficial, demarcated lesion following the lymphatic chain	Erysipelas (GAS)
Well infant, fever, no focus	UTI → urine analysis
Adolescent girl with UTI, gram-positive	Staphylococcus saprophyticus
Hemorrhagic cystitis in an immunocompromised child	Adenovirus
Pyuria with negative culture	TB (sterile pyuria)
UTI in a neonate	Group B streptococcus
Candida in urine with a catheter	Remove the catheter
Nitrite positive	High specificity (98%)
LE or nitrite positive	Highest sensitivity (93%)

Case-Based Practice MCQs

Cases 1–5 are the lecture's own cases; 6–14 are written from the lecture content only. Answers are on the next page so you can test yourself first.

1. A 3-year-old has low-grade fever, runny nose, cough and sore throat. What is the most appropriate management?
 - A. Penicillin V for 10 days
 - B. Supportive treatment
 - C. Rapid antigen test and amoxicillin if positive
 - D. Clindamycin
2. A 6-year-old has high fever and a severe sore throat with no cough and tonsillar exudate. What is the main reason to treat with antibiotics?
 - A. To shorten the viral illness
 - B. To prevent acute rheumatic fever
 - C. To prevent sinusitis only
 - D. To treat EBV
3. A 6-month-old infant has fever, vomiting and excessive crying. Ear examination shows a bulging tympanic membrane. What is the best management?
 - A. Observe for 48–72 h
 - B. Antibiotics now, for 10 days
 - C. Antibiotics for 5 days
 - D. Tympanocentesis first
4. A 6-year-old boy has redness and swelling of the chin with golden crusts. He looks well and is afebrile. What is the best treatment?
 - A. Topical mupirocin
 - B. IV flucloxacillin
 - C. Surgical drainage
 - D. Anti-pseudomonal antibiotic
5. A 9-month-old has fever for three days, looks well, T 38.9 °C, no focus. What is the next step?
 - A. Chest X-ray
 - B. Urine analysis
 - C. Throat culture
 - D. Reassure and discharge
6. Which statement about the rapid antigen detection test for GAS is correct?
 - A. Sensitivity 97%, so a negative result excludes GAS
 - B. High specificity (> 95%) but lower sensitivity, so a negative result needs a throat culture
 - C. It takes 2–3 days
 - D. It is the gold standard
7. A 3-year-old with AOM received amoxicillin 3 weeks ago. What is the first-line antibiotic now?
 - A. High-dose amoxicillin
 - B. High-dose amoxicillin-clavulanate
 - C. IV ceftriaxone
 - D. Azithromycin
8. Which is the hallmark sign of AOM?
 - A. Redness of the TM
 - B. Bulging TM
 - C. Ear tugging
 - D. Fever
9. A child has nasal discharge and cough for 12 days, not improving, with no fever. What is this?
 - A. Viral URTI, reassure
 - B. Persistent acute bacterial sinusitis
 - C. Severe acute bacterial sinusitis
 - D. Needs a sinus CT
10. A neutropenic child with leukemia develops a skin lesion with a necrotic black center. What is the most likely organism?
 - A. *S. aureus*
 - B. *S. pyogenes*
 - C. *Pseudomonas aeruginosa*
 - D. Group B streptococcus
11. A 10-month-old is not toilet trained. Which urine collection method and cut-off are recommended?
 - A. Urine bag, any growth
 - B. Transurethral catheter, > 50,000 CFU/mL
 - C. Midstream, > 100,000 CFU/mL
 - D. Suprapubic aspiration, > 100,000 CFU/mL
12. A 4-year-old with a febrile UTI is well, tolerating oral intake. What is the correct statement?
 - A. IV antibiotics are more effective than oral
 - B. Oral and parenteral starts are equally effective in children ≥ 2 months
 - C. Ampicillin is the preferred choice in Saudi children
 - D. Co-trimoxazole is first line
13. A 1-year-old with a first febrile UTI has recovered. When should RBUS be done?
 - A. Immediately, during the fever
 - B. 2 to 6 weeks after the acute infection
 - C. Only if a second UTI occurs
 - D. Never under 3 years
14. A urinalysis shows pyuria but repeated cultures are sterile. What should be considered?
 - A. *E. coli*
 - B. Adenovirus
 - C. Tuberculosis
 - D. Candida

Answer Key

1	B: most likely viral (age ≤ 3 , cough, runny nose) → supportive treatment.
2	B: GAS pharyngitis is treated to prevent rheumatic fever (9-day window). Per the mind-map, antibiotics cannot prevent glomerulonephritis.
3	B: age < 2 y (especially < 6 m) must be treated; < 2 y gets 10 days to prevent treatment failure.
4	A: well and afebrile → topical antibiotic (mupirocin); ill and febrile → systemic (e.g. flucloxacillin).
5	B: in infants, fever without focus is often the only sign of UTI → urine analysis.
6	B: RADT sensitivity 70–97%, specificity $> 95\%$; negative → wait for throat culture (97% / 99%, takes 2–3 days).
7	B: amoxicillin within 30 days (or purulent conjunctivitis) → high-dose amox/clav, because of H. influenzae resistance.
8	B: bulging TM is the hallmark; redness can occur without acute suppurative OM.
9	B: persistent = nasal discharge and/or cough > 10 days and not improving; no imaging needed if uncomplicated.
10	C: ecthyma gangrenosum → anti-pseudomonal therapy and blood culture.
11	B: transurethral catheter is first choice for non-toilet-trained children, $> 50,000$ CFU/mL (or $> 10,000$ with fever and pyuria). Bags are not recommended.
12	B: oral = parenteral in children ≥ 2 months unless toxic or unable to take orally. Saudi data: ~70% ampicillin and ~60% co-trimoxazole resistance.
13	B: 2–6 weeks after the acute phase avoids false positives from inflammation. Early US only if severely ill, complicated, obstructed, or not improving within 48 h.
14	C: TB causes sterile pyuria ($\uparrow$ WBC in urine, culture –ve).

⚠ Inconsistencies in the Source (check with the lecturer)

Nothing below was changed in the summary. Everything was kept as written in the source and flagged, so you can decide which version your exam follows.

#	Where	The conflict
1	Pharyngitis age	The mind-map says bacterial is mainly > 4 y and viral < 4 y. The Case 1/2 notes use a cut-off of 3 y. The slide table says the bacterial peak is 5–11 y, and Centor gives a point for 3–14 y.
2	Glomerulonephritis	The Case 2 note says antibiotics are given "to prevent complication of rheumatic fever and glomerulonephritis". The mind-map crosses out glomerulonephritis and notes (Arabic) that antibiotics can't prevent it. The mind-map version matches standard teaching.
3	Case 1 note	"Swelling in the tonsils + headache + unilateral tenderness in cervical LN" is written beside the viral Case 1 photo. On the slide's own table these are bacterial features, so the note may belong to Case 2.
4	AOM H. influenzae	The note says "nontypeable, mostly type B". Those are contradictory: nontypeable strains have no capsule, so they cannot be type b. The typed box also lists "H. influenzae type B" as the AOM organism. Nontypeable H. influenzae is the usual AOM strain.
5	AOM ceftriaxone	The flowchart reads "IV Ceftriaxone 50 mg/d daily for 3 days". The unit is almost certainly meant to be 50 mg/kg/day. Please confirm.
6	ABS diagnosis note	"First wait 48 h then give tx" disagrees with the ABS flowchart, which says outpatient observation for 3 days for non-severe cases.
7	Erysipelas duration	The slide says oral anti-streptococcal therapy for 7–14 days. The handwritten note on the photo slide says cellulitis and erysipelas both need oral antibiotics for 5–7 days.
8	Circumcision	The note explains the 8-fold UTI reduction "because of short urethra". Circumcision does not change urethral length; the protective effect is usually attributed to removing the foreskin, where bacteria colonize. The "short urethra" reason explains why girls get more UTIs.
9	Cystitis duration	The handwriting says "less than 5 d". The mind-map says 3–5 days. Table 3 says 3–7 days, and the typed notes say 3–7 days, extended to 10.
10	Urine bag	The slide says the bag is "helpful mainly in children with a negative culture result to exclude UTI". The typed note says a bag urinalysis "rules out UTI". Both mean a negative bag result can help exclude UTI, but a positive bag culture must never be used for diagnosis.
11	Oral cephalosporin	The handwriting says "2nd or 3rd generation cephalosporin". The table and typed notes list cephalexin (1st gen), cefuroxime and cefprozil (2nd gen) as first line, with cefixime (3rd gen) only as the alternative.
12	Recurrent UTI	The handwriting says "more than 2 UTI in 1 year". The slide defines recurrent UTI as two or more episodes within 12 months.
13	RBUS timing	The mind-map says "after 6 weeks". The slides say 2 to 6 weeks after the acute infection.

Also not recoverable from the source: one handwritten line under the sinus anatomy figure and one under "Ecthyma gangrenosum" on the photo slide are cut off at the slide edge (the latter continues on the next slide and is included). The lecture has no ABS complications slide, and nothing on erythrasma beyond its photo label.